



LT10b – Stack ADT

ABSTRACT DATA TYPE

What is a Stack ?

1. A stack has only one opening at the top.
2. "**POP**" when you take out from a stack and "**PUSH**" when put into a stack.
3. First item will be at the bottom and the last item will be at the top.
4. A stack has a Last-In-First-Out (LIFO) property.
5. Cannot "POP" an empty stack.



Using Python List for the Stack ADT

make_empty_stack() : returns an empty list

push(stack, item) : append an element into the list

pop(stack) : removes the last element in the list and returns the removed element

peek(stack) : returns the last element in the list but does not remove it from the list

Stack operations

```
s = make_empty_stack()
```

```
pop(s) → None : empty stack, nothing to pop
```

```
push(s, 7)
```

```
push(s, 5)
```

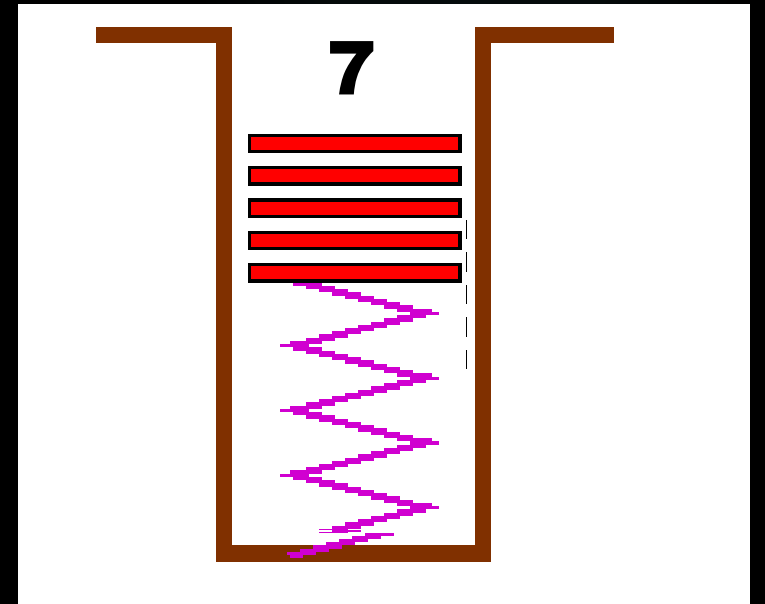
```
push(s, 3)
```

```
pop(s) → 3
```

```
pop(s) → 5
```

```
is_empty(s) → False
```

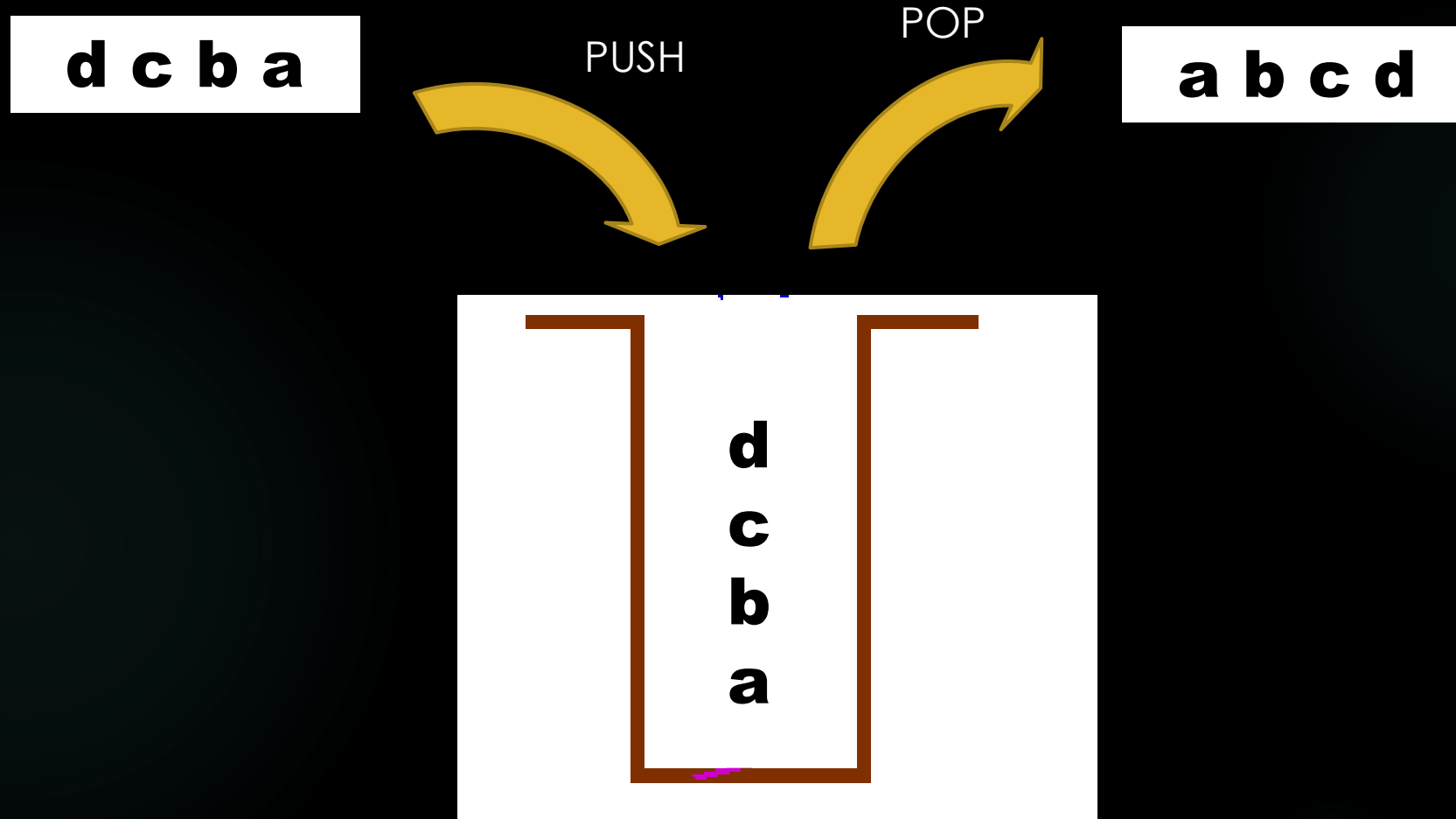
```
peek(s) → 7
```





LT10b – Stack ADT Applications

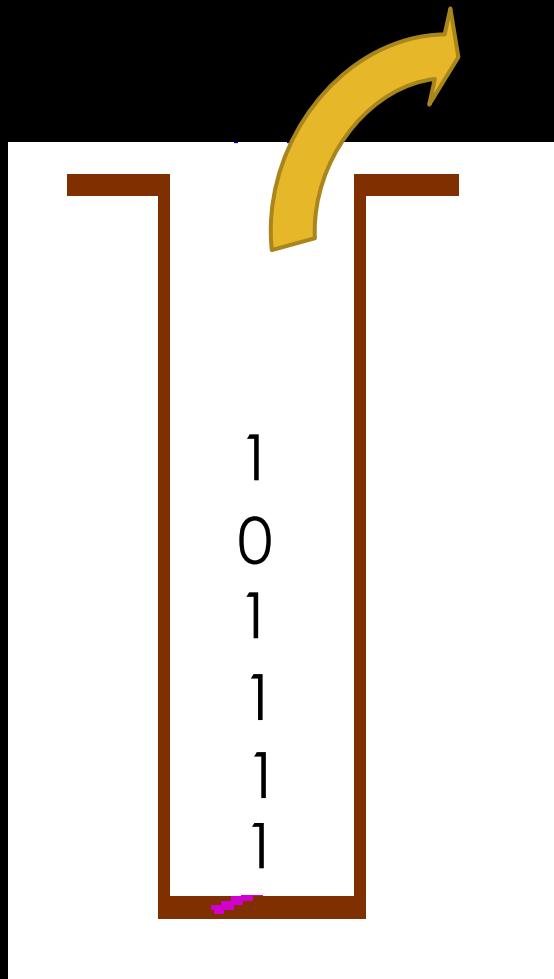
1. Reversing a Sequence



2. Converting Denary to Binary

- ▶ Use successive division by 2 until the quotient is 0:
- ▶ The remainders form the answer:

2	47	
2	23	→ 1
2	11	→ 1
2	5	→ 1
2	2	→ 1
2	1	→ 0
	0	→ 1



$$47_{10} = 101111_2$$

3. Checking for Balanced Parentheses

([] ({ () })) is Balanced **(({ } []))** is Not

Simple counting is not enough to check for balance.

You can do it with a stack. Going from left to right ...

1. If you see a '(', '[', or '{', push it into the stack;
2. If you see a ')', ']', or '}', pop an element out of the stack and check whether it is a corresponding '(', '[', or '{'.

When you reach the end, check that the stack is empty.

Arithmetic Notation for Calculation

- ▶ Infix notation : the operator is written in between the operands.

Eg : $A + B$

- ▶ Prefix notation : the operator is written before the operands.

Eg : $+ A B$

- ▶ Postfix notation : the operator is written after the operands.

Eg : $A B +$

Computing with Postfix Notation

► Postfix notation : $A B +$

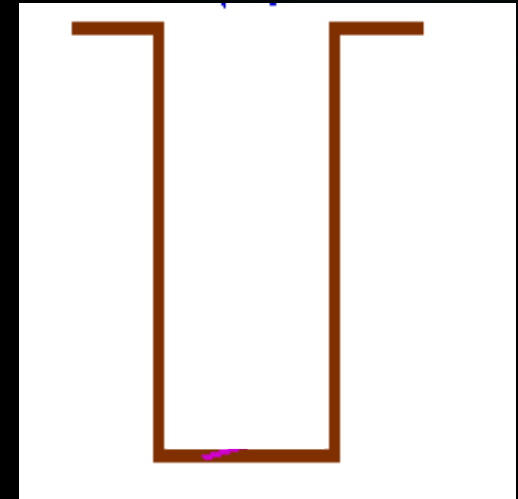
Is this $A + B$ or $B + A$?

↓ ↓ ↓
 $A B -$

$A - B$

► Postfix notation : $A B -$

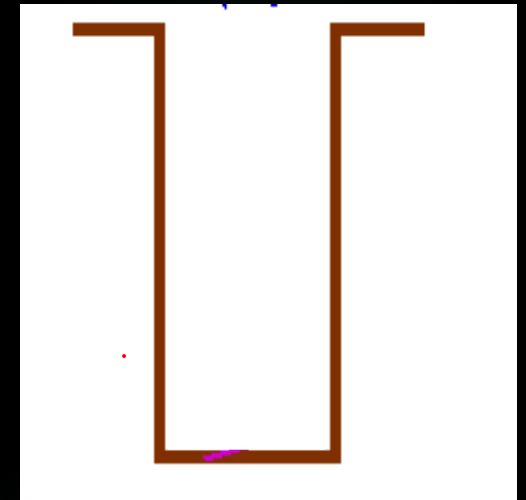
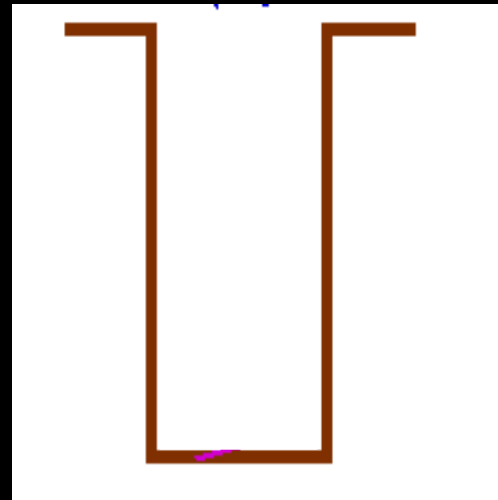
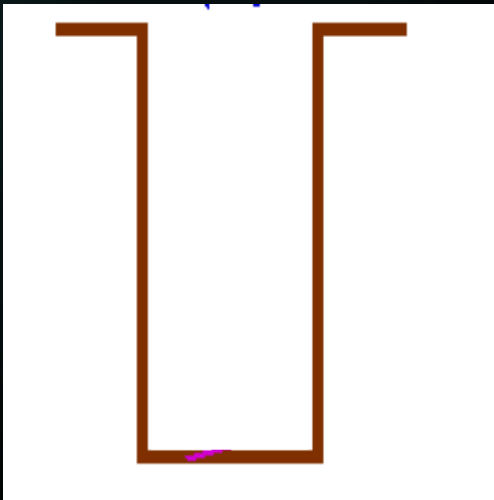
Is it $A - B$ or $B - A$?



4. Computing with Postfix Notation

► Postfix notation : **3 4 * 5 +**

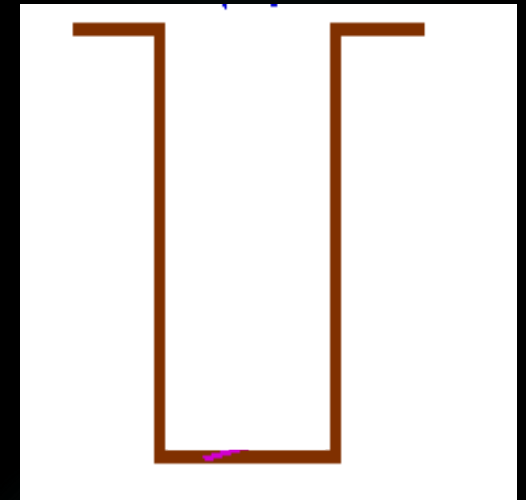
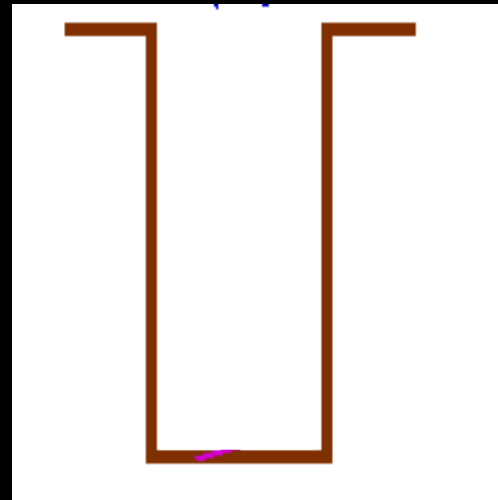
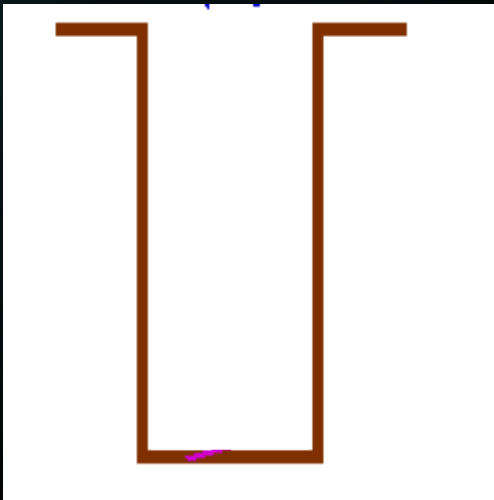
► Infix notation : **3 * 4 + 5**



4. Computing with Postfix Notation

► Postfix notation : **3 4 5 * +**

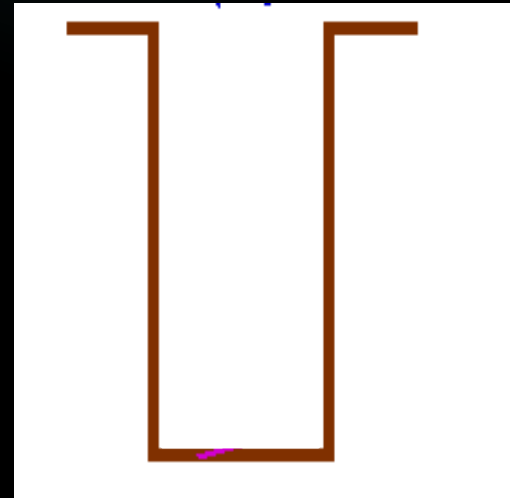
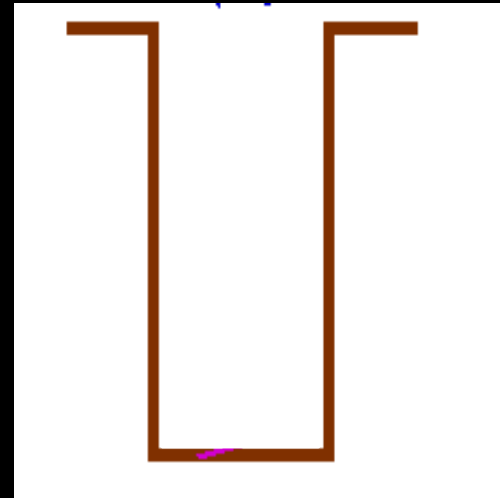
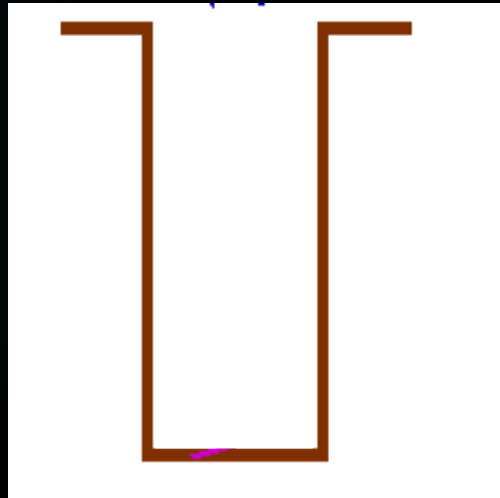
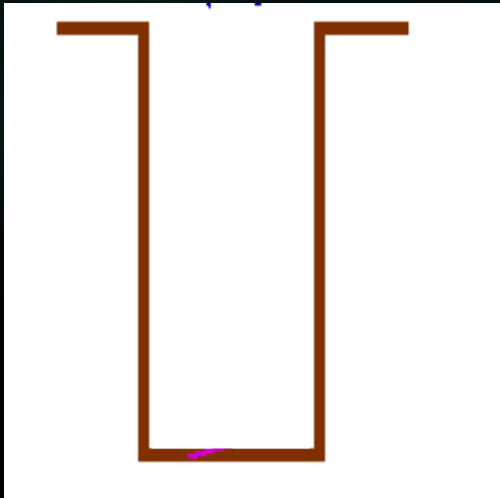
► Infix notation : **3 + 4 * 5**



4. Computing with Postfix Notation

► Postfix notation : **3 4 5 + * 2 /**

► Infix notation : **3 * (4 + 5) / 2**



The End